

Data Structures II

Assignment 1

A college stores student roll numbers and frequently performs search, insertion, and deletion.

Question:

Explain how BST can be used to manage the records. Construct the BST for **50, 30, 70, 20, 40** and perform deletion of 30.

A game stores player scores and requires fast searching.

Question:

Explain why AVL Tree is preferred over BST. Construct AVL Tree for **10, 20, 30, 40, 50** and show rotations.

A database stores large records and requires efficient disk access.

Question:

Explain how B-Tree is used for indexing. Insert keys **10, 20, 5, 6, 12, 30, 7, 17** in a B-Tree of order 3.

Books are stored using IDs and often accessed in ranges.

Question:

Construct a B+ Tree for **5, 10, 15, 20, 25, 30** and explain leaf node linking.